

CAPTIONING: EVERYTHING, EVERYWHERE, ALL AT ONCE?

Integrating Automated Speech Recognition into a Digital Preservation Workflow

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Introduction

In early 2026, we were tasked with creating and preserving captions and transcripts for all AV material in our digital preservation workflow. This poster describes our process transforming a small pilot project into an ongoing service, challenges encountered in scaling up, and lessons learned for future integrations of automated tools.

Scripts and Software

run_whisper.py - finds AV files, builds file list, and launches whispervt.py

whispervtt.py - creates caption and transcript files with FADGI metadata

move_vtts.py - moves processed AV and captions/transcript files to synced folders

webvtt_metadata.py - creates or revises FADGI metadata for captions/transcripts

whisperGUI.py - graphic interface for creating caption and transcript files

OneDrive: syncs files between workstations

Zapier: syncs Airtable records with OneDrive

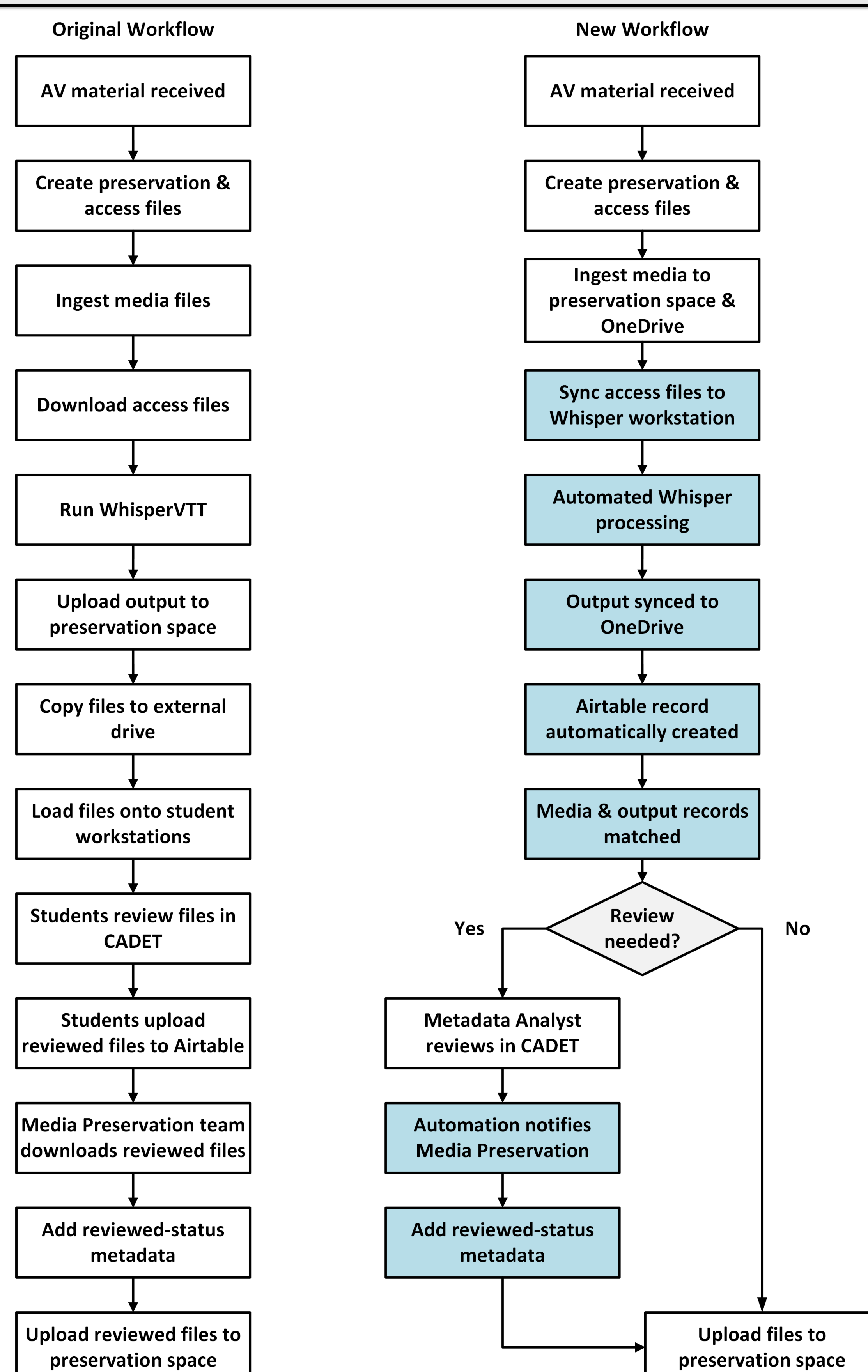
Airtable: matches media files with caption files; notifies staff when files are ready for review/upload

Conclusions

In considering future integrations of automated tools, we would emphasize lowering skill barriers to conducting this work, prioritizing time and resources according to user and organizational needs, and metadata documentation of the use of AI tools and level of human review of AI-generated resources.

Original workflow: captioning work took place downstream and separate from the preservation workflow; file copying and transfers were initiated manually; students reviewed and corrected all caption files created by Whisper.

This plan worked for a small volume of material, but to caption all digital AV in special collections, we faced the challenges of an increased workload and limited resources, and needed a more efficient approach. We changed where our captioning workflow intersected with our preservation workflow, automated steps where possible, and developed a framework to guide the allocation of reviewer time.



New workflow: when AV access files are created, they are synced to the captioning workstation, which enables captions to be created concurrently with preservation ingest; all AV is included by default, eliminating the selection step; tasks are automated where possible, so the workflow proceeds with minimal staff intervention; caption reviewer time is prioritized to patron request items and output that is likely to be less accurate, based on the AV content’s characteristics and Whisper’s strengths and weaknesses.

Self-service workflow: we created a GUI for libraries at Emory to run Whisper themselves without needing command-line skills; this reduces the volume of material entering the main workflow.

These changes have introduced a new challenge of increased complexity, with more tools, scripts, and automations to keep in sync, but have overall allowed us to successfully scale up our work.

Statistics

Old workflow, Sep 2023 - Dec 2025:

753 files (453 content hours) created
409 files (229 content hours) reviewed

New workflow, Jan 2026 - present:

1066 files (766 content hours) created
171 files (61 content hours) reviewed

Python scripts

github.com/ninarao
whispervtt
webvtt_metadata
WhisperGUI



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